

GAS DYNAMICS

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Gas Kinetics

- Molecular motion, interactions, statistical behaviour
- Kinetic Theory of Gases
- Mean free path, collision frequency

Knudsen number Kn : ratio of mean free path λ to the characteristic length L of the corresponding flow field.

$$Kn > 0.1$$

Gas Dynamics

- Macroscopic gas flows, pressure temperature velocity changes
- Continuum Mechanics, compressible flow, Euler, Navier-Stokes
- Reynolds number, Mach number

$$Kn < 0.01$$

Entropy of a gas $s_2 - s_1 = C_p \ln \frac{T_2}{T_1} - R \ln \frac{p_2}{p_1}$

Mayer's Relation $C_p - C_v = R$

Speed of sound $a = \sqrt{\gamma RT}$

Isentropic relations

$$P V^\gamma = \text{constant}$$

$$\frac{T_2}{T_1} = \left(\frac{p_2}{p_1} \right)^{\frac{\gamma}{\gamma-1}}$$

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Isentropic flow

Perfect gas : $P V = n R T$

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Calorically perfect $\rightarrow C_p, C_v$ - constant with respect to temperature T

Thermally perfect $\rightarrow C_p = f(T), C_v = f(T)$
Chemically perfect $\rightarrow$ and considering chemical reactions

Classius inequality $ds > \frac{\delta q}{T} \geq \frac{\delta q}{T}$

$$S_2 - S_1 = C_p \ln \left(\frac{T_2}{T_1} \right) - R \ln \left(\frac{P_2}{P_1} \right)$$

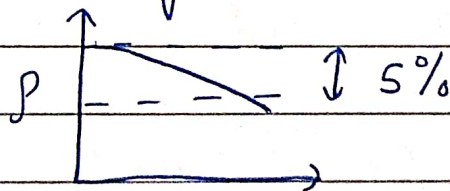
$$S_2 - S_1 = C_v \ln \left(\frac{T_2}{T_1} \right) + R \ln \left(\frac{V_2}{V_1} \right)$$

for a typical reversible adiabatic system that is (isentropic) $ds = 0$

Isentropic relations $\frac{T_2}{T_1} = \left(\frac{P_2}{P_1} \right)^{\frac{\gamma-1}{\gamma}} = \left(\frac{P_2}{P_1} \right)^{\gamma-1}$

Compressible flow : change in density

$$\frac{\rho}{\rho_1}$$



Pressure or Mach Number

$$dp = \tau \rho dP$$

$\downarrow$

compressibility factor (inverse of bulk modulus)

• Adiabatic Energy Equation

from steady flow energy equation, $h_1 + \frac{1}{2} V_1^2 = h_2 + \frac{1}{2} V_2^2$

For a typical non isentropic adiabatic flow,
Total condition does not
remain constant



Any flow, stagnation enthalpy $h_{01} = h_{02}$ and
stagnation temperature $T_{01} = T_{02}$. These conditions
won't change but stagnation pressure will vary.

For isentropic reversible adiabatic,
 $h_{01} = h_{02}$, $T_{01} = T_{02}$, $P_{01} = P_{02}$
stagnation conditions remain constant

Stagnation case: $h_1 + \frac{1}{2} V_1^2 = h_2 + \frac{1}{2} V_2^2$
 $h_1 + \frac{1}{2} V_1^2 = h_0$ (stagnation enthalpy)

• Adiabatic Energy Equation in terms of sound velocity
 $h + \frac{1}{2} V^2 = h_0$

$$C_p T + \frac{V^2}{2} = C_p T_0$$

$$T_0 = T + \frac{V^2}{2 C_p}$$

(dividing by C_p)

$$\frac{T_0}{T} = 1 + \frac{V^2}{2 C_p T}$$

(dividing by T)

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$$\text{Since } C_p = \frac{\gamma R}{\gamma - 1}, \quad \frac{T_0}{T} = 1 + \frac{V^2}{2 \frac{\gamma R}{\gamma - 1} T}$$

$$\frac{T_0}{T} = 1 + \frac{(\gamma - 1) V^2}{2 a^2}$$

$$\frac{T_0}{T} = 1 + \left(\frac{\gamma - 1}{2}\right) M^2$$

$$\frac{P_0}{P} = \left[1 + \left(\frac{\gamma - 1}{2}\right) M^2 \right]^{\frac{\gamma}{\gamma - 1}}$$

$$\frac{P_0}{P} = \left[1 + \left(\frac{\gamma - 1}{2}\right) M^2 \right]^{\gamma - 1}$$

sound velocity $a = \sqrt{\gamma R T}$

$$a^2 = \gamma R T = \gamma \frac{P}{\rho}$$

$$h + \frac{1}{2} V^2 = h_0 = \text{constant}$$

$$C_p T + \frac{V^2}{2} = \text{constant}$$

$$\frac{\gamma R}{\gamma - 1} T + \frac{V^2}{2} = \text{constant}$$

$$\frac{a^2}{\gamma - 1} + \frac{V^2}{2} = \text{constant}$$

$$h_0 = C_p T_0 = \frac{\gamma R T_0}{\gamma - 1} = \frac{a_0^2}{\gamma - 1}$$

$$\therefore \frac{a^2}{\gamma - 1} + \frac{V^2}{2} = \frac{a_0^2}{\gamma - 1} = \text{constant}$$

- Critical velocity of sound (v^* or a^*)
 $T = T^*$ temperature when Mach = 1

$$\begin{aligned}\frac{a^{*2}}{\gamma-1} + \frac{v^{*2}}{2} &= a^{*2} \left(\frac{2+\gamma-1}{2(\gamma-1)} \right) \\ &= \frac{a^{*2}}{2} \left(\frac{\gamma+1}{\gamma-1} \right) \\ &= \frac{a^2}{\gamma-1} + \frac{v^2}{2} = \frac{a_0^2}{\gamma-1}\end{aligned}$$

- Maximum velocity when $T = 0K$ (V_{max}) (expanding gas to vacuum)
 $\frac{V_{max}^2}{2} = \text{constant}$

$$\frac{a^2}{\gamma-1} + \frac{v^2}{2} = \frac{V_{max}^2}{2} = \frac{a^*}{2} \left(\frac{\gamma+1}{\gamma-1} \right) = \frac{a_0^2}{\gamma-1}$$

- Bernoulli's relation for a typical compressible flow
 $h + \frac{1}{2}v^2 = h_0$

$$C_p T + \frac{1}{2}v^2 = C_p T_0$$

$$\frac{\gamma R T}{\gamma-1} + \frac{1}{2}v^2 = \frac{\gamma R T_0}{\gamma-1}$$

$$\frac{\gamma}{\gamma-1} \left(\frac{P}{\rho} \right) + \frac{v^2}{2} = \frac{\gamma}{\gamma-1} \left(\frac{P_0}{\rho_0} \right)$$

$$\frac{P}{\rho} = \frac{P}{\rho_0} \frac{\rho_0}{\rho} \frac{P_0}{P} = \frac{P}{\rho_0} \frac{P_0}{P} \left(\frac{P}{P_0} \right)^{-\frac{1}{\gamma}} = \left(\frac{P}{P_0} \right)^{\frac{\gamma-1}{\gamma}} \frac{P_0}{\rho_0}$$

$$\frac{\gamma}{\gamma-1} \left(\frac{P}{P_0} \right)^{\frac{\gamma-1}{\gamma}} \frac{P_0}{\rho_0} + \frac{v^2}{2} = \frac{\gamma}{\gamma-1} \frac{P_0}{\rho_0}$$

28/3. Critical Parameters - P^* , T^* , ρ^* , v^*

$$h + \frac{1}{2} v^2 = h_0$$

$$\frac{a^2}{\gamma-1} + \frac{v^2}{2} = \frac{a^{*2}}{2} \left(\frac{\gamma+1}{\gamma-1} \right) = \frac{a_0^2}{\gamma-1} = \frac{v_{max}^2}{2}$$

Stagnation Static Conditions $\frac{P_0}{P} = \left(\frac{T_0}{T} \right)^{\frac{\gamma}{\gamma-1}} = \left(1 + \frac{\gamma-1}{2} M^2 \right)^{\frac{\gamma}{\gamma-1}}$

$$\frac{P_0}{P} = \left(1 + \frac{\gamma-1}{2} M^2 \right)^{\frac{\gamma}{\gamma-1}}$$

Critical Parameters when $M=1$,

$$P = P^*, T = T^*, \rho = \rho^*, v = v^*$$

$$\frac{T^*}{T_0} = \frac{1}{1 + \frac{\gamma-1}{2}} = \frac{2}{\gamma+1}$$

$$\frac{P^*}{P_0} = \left(\frac{2}{\gamma+1} \right)^{\frac{\gamma}{\gamma-1}}$$

$$\frac{\rho^*}{\rho} = \left(\frac{2}{\gamma+1} \right)^{\frac{1}{\gamma-1}}$$

For air, $\gamma = 1.4$

$$\frac{T^*}{T_0} = 0.833$$

$$\frac{P^*}{P_0} = 0.528$$

$$\frac{\rho^*}{\rho_0} = 0.834$$

condition to check for sonic condition
(happens at throat)

Compressibility with respect to pressure coefficient

$$\frac{P_0}{P} = \left(1 + \frac{\gamma-1}{2} M^2 \right)^{\frac{\gamma}{\gamma-1}}$$

$$C_p = \frac{P_0 - P}{\frac{1}{2} \rho v^2}$$

By Binomial expansion,

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!} x^2 + \frac{n(n-1)(n-2)}{3!} x^3 + \dots$$

$$n = \frac{\gamma}{\gamma-1} \quad \gamma = \frac{\gamma-1}{2} M^2$$

$$\frac{P_0}{P} = 1 + \frac{\gamma}{2} M^2 + \frac{\gamma}{\gamma-1} \left(\frac{\gamma}{\gamma-1} - 1 \right) \left(\frac{\gamma-1}{2} M^2 \right)^2 \times \frac{1}{2!} \\ + \frac{\gamma}{\gamma-1} \left(\frac{\gamma}{\gamma-1} - 1 \right) \left(\frac{\gamma}{\gamma-1} - 2 \right) \frac{1}{3!} \left(\frac{\gamma-1}{2} M^2 \right)^3 \dots$$

$$\frac{P_0}{P} = 1 + \frac{\gamma}{2} M^2 + \frac{\gamma}{2} (\gamma - \gamma + 1) \frac{M^4}{4} + \frac{\gamma}{6} (\gamma - \gamma + 1) (\gamma - 2\gamma + 2) \times \frac{M^6}{8} \dots$$

$$\frac{P_0}{P} = 1 + \frac{\gamma}{2} M^2 + \frac{\gamma}{8} M^4 + \frac{\gamma(2-\gamma)}{48} M^6 + \dots$$

$$\frac{P_0}{P} - 1 = \frac{\gamma}{2} M^2 + \frac{\gamma}{8} M^4 + \frac{\gamma(2-\gamma)}{48} M^6 + \dots$$

$$\frac{P_0 - P}{P} = \frac{\gamma}{2} M^2 \left(1 + \frac{M^2}{4} + \frac{2-\gamma}{24} M^4 + \dots \right)$$

$$\frac{P_0 - P}{P \left(\frac{\gamma}{2} M^2 \right)} = 1 + \frac{M^2}{4} + \frac{2-\gamma}{24} M^4 + \dots$$

$$\frac{\gamma}{2} M^2 = \frac{\gamma}{2} \frac{V^2}{a^2} = \frac{\gamma}{2} \frac{V^2}{\gamma R T} = \frac{V^2}{2 R T}$$

$$C_p = \frac{P_0 - P}{P \cdot \frac{V^2}{2 R T}} = \frac{P_0 - P}{\frac{1}{2} P V^2} \\ = 1 + \frac{M^2}{4} + \frac{2-\gamma}{24} M^4 + \dots$$

when $M = 0.4$, $C_p = 1.04064$

when $M = 0.6$, $C_p = 1.0934$

when $M = 0.8$, $C_p = 1.17024$

when $M = 1$, $C_p = 1.275$

As M increases, C_p increases

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Characteristic Mach Number M^* (NOT CRITICAL)

$$M^* = \frac{V}{a^*}$$

To prove: As $M \rightarrow \infty$, $M^* \rightarrow$ finite value

$$\frac{a^2}{\gamma-1} + \frac{V^2}{2} = \frac{a^{*2}}{2} \left(\frac{\gamma+1}{\gamma-1} \right)$$

$$\begin{aligned} \text{Dividing by } V^2, \quad \frac{a^{*2}}{2V^2} \frac{\gamma+1}{\gamma-1} &= \frac{a^2/V^2}{\gamma-1} + \frac{1}{2} \\ &= \frac{1/M^2}{\gamma-1} + \frac{1}{2} \end{aligned}$$

$$\frac{a^{*2}}{2V^2} = \left(\frac{1}{M^2(\gamma-1)} + \frac{1}{2} \right) \left(\frac{\gamma-1}{\gamma+1} \right)$$

$$\frac{1}{2M^{*2}} = \frac{2 + (\gamma-1)M^2}{2(\gamma+1)M^2} \left(\frac{\gamma-1}{\gamma+1} \right)$$

$$\frac{1}{M^{*2}} = \frac{2 + (\gamma-1)M^2}{(\gamma+1)M^2}$$

$$M^{*2} = \frac{(\gamma+1)M^2}{2 + (\gamma-1)M^2}$$

as $M \rightarrow \infty$ and $\gamma = 1.4$

$$M^{*2} = \frac{M^2(\gamma+1)}{M^2 \left(\frac{2}{M^2} + (\gamma-1) \right)} = \frac{\gamma+1}{\gamma-1}$$

$$M^* = \sqrt{\frac{\gamma+1}{\gamma-1}} \quad (\text{finite value})$$

Area velocity relations

flow through a varying area duct - quasi 1D flow

$\Rightarrow A = A(x)$ area is a function of only 1 direction

$\rho AV = \text{constant}$

$$d(\rho AV) = 0$$

$$\frac{dp}{\rho} + \frac{dA}{A} + \frac{dv}{v} = 0 \quad (\text{Euler's relation})$$

$$dp = -\rho v dv$$

$$\frac{dp}{\rho} = -v dv$$

$$= \frac{dp}{\rho} \times \frac{dp}{dp}$$

$$-v dv = a^2 \times \frac{dp}{\rho}$$

$$-v^2 \frac{dv}{v} = a^2 \frac{dp}{\rho}$$

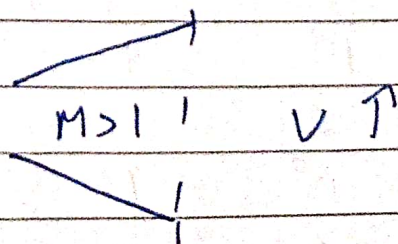
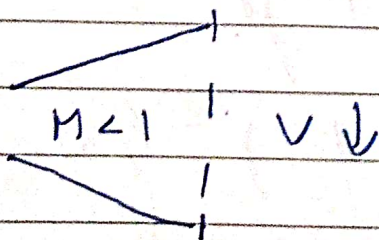
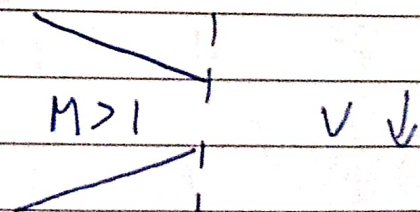
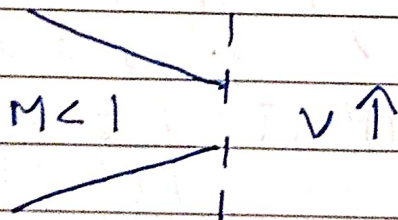
$$\frac{dp}{\rho} = -M^2 \frac{dv}{v}$$

$$-M^2 \frac{dv}{v} + \frac{dA}{A} + \frac{dv}{v} = 0$$

$$\frac{dA}{A} = (M^2 - 1) \frac{dv}{v}$$

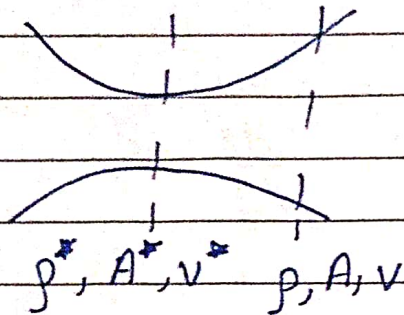
for subsonic flow, as area increases, velocity decreases

for supersonic flow, as area increases, velocity increases



$M=1, v = \text{constant}$
(Throat)

Area-Mach Number Relations



$$\begin{aligned} \rho^* A^* v^* &= \rho A v \\ \frac{A}{A^*} &= \frac{\rho^* v^*}{\rho v} \\ &= \frac{\rho}{\rho_0} \frac{\rho_0}{\rho} \frac{v^*}{v} \end{aligned}$$

$$\frac{\rho^*}{\rho_0} = \left(\frac{2}{\gamma+1} \right)^{\frac{1}{\gamma-1}}$$

$$\frac{\rho_0}{\rho} = \left(1 + \left(\frac{\gamma-1}{2} \right) M^2 \right)^{\frac{1}{\gamma-1}}$$

$$M^* = \frac{v}{a^*}$$

$$v = v^*$$

$$\frac{v^*}{v} = \frac{1}{M^*}$$

$$= \sqrt{\frac{2 + (\gamma-1) M^2}{(\gamma+1) M^2}}$$

$$\frac{A}{A^*} = \left(\frac{2}{\gamma+1} \right)^{\frac{1}{\gamma-1}} \left(1 + \left(\frac{\gamma-1}{2} \right) M^2 \right)^{\frac{1}{\gamma-1}} \sqrt{\frac{2 + (\gamma-1) M^2}{(\gamma+1) M^2}}$$

$$\frac{A}{A^*} = \left(\frac{2}{\gamma+1} \right)^{\frac{1}{\gamma-1}} \left(1 + \frac{\gamma-1}{2} M^2 \right)^{\frac{1}{\gamma-1}} \frac{1}{M} \left[\frac{2 + \frac{\gamma-1}{2} M^2}{\gamma+1} \right]^{\frac{1}{2}}$$

$$\frac{A}{A^*} = \frac{1}{M} \left(\frac{2}{\gamma+1} \left(1 + \frac{\gamma-1}{2} M^2 \right) \right)^{\frac{1}{\gamma-1}} \left[\frac{2 + \frac{\gamma-1}{2} M^2}{\gamma+1} \right]^{\frac{1}{2}}$$

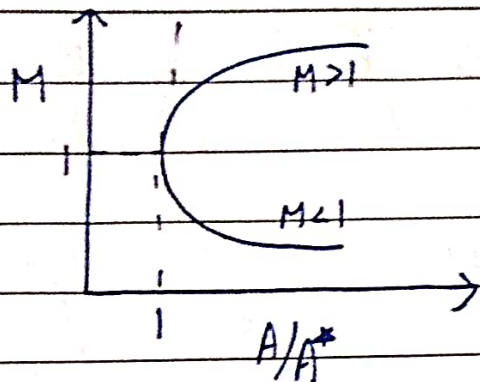
$$\frac{A}{A^*} = \frac{1}{M} \left(\frac{2}{\gamma+1} \left(\frac{2 + (\gamma-1) M^2}{2} \right) \right)^{\frac{1}{\gamma-1}} \left[\frac{2 + \frac{\gamma-1}{2} M^2}{\gamma+1} \right]^{\frac{1}{2}}$$

$$\frac{A}{A^*} = \frac{1}{M} \left(\frac{2}{\gamma+1} + \frac{\gamma-1}{\gamma+1} M^2 \right)^{\frac{1}{\gamma-1}} \left[\frac{2 + \frac{\gamma-1}{2} M^2}{\gamma+1} \right]^{\frac{1}{2}}$$

$$\frac{A}{A^*} = \frac{1}{M} \left(\frac{2}{\gamma+1} + \frac{\gamma-1}{\gamma+1} M^2 \right)^{\frac{1}{\gamma-1} + \frac{1}{2}}$$

$$\frac{A}{A^*} = \frac{1}{M} \left(\frac{2}{\gamma+1} + \frac{\gamma-1}{\gamma+1} M^2 \right)^{\frac{2+\gamma-1}{2(\gamma-1)}}$$

$$\frac{A}{A^*} = \frac{1}{M} \left(\frac{2}{\gamma+1} + \frac{\gamma-1}{\gamma+1} M^2 \right)^{\frac{\gamma+1}{2(\gamma-1)}}$$



$\frac{A}{A^*}$ cannot be less than 1

$\frac{A}{A^*} > 1$ $M \rightarrow$ subsonic
 $M \rightarrow$ supersonic

1/4 . Mass flow rate : $\frac{\dot{m} \sqrt{T_0}}{A P_0}$

$$\dot{m} = \rho^* A^* v^*$$

Dividing by A , $\frac{\dot{m}}{A} = \rho^* \frac{A^*}{A} v^*$

$$\rho^* = \frac{P^*}{RT^*}$$

$$v^* = \sqrt{\gamma RT^*}$$

$$\frac{\dot{m}}{A} = \frac{P^*}{RT^*} \sqrt{\gamma RT^*} \frac{A^*}{A}$$

$$\frac{\dot{m}}{A} = \frac{P^*}{\sqrt{T^*}} \sqrt{\frac{\gamma}{R}} \frac{A^*}{A}$$

$$\frac{\dot{m}}{A} \sqrt{\frac{R}{\gamma}} = \frac{P^*}{\sqrt{T^*}} \frac{A^*}{A}$$

$$\frac{P^*}{P_0} = \left(\frac{2}{\gamma+1} \right)^{\frac{\gamma}{\gamma-1}}$$

$$\frac{T^*}{T_0} = \frac{2}{\gamma+1}$$

$$\frac{\dot{m}}{A} \sqrt{\frac{R}{\gamma}} = \frac{P_0 \left(\frac{2}{\gamma+1} \right)^{\frac{\gamma}{\gamma-1}}}{\left(T_0 \left(\frac{2}{\gamma+1} \right) \right)^{1/2}} \cdot \frac{A^*}{A}$$

$$\frac{\dot{m}}{A} \sqrt{\frac{R}{\gamma}} = \frac{P_0}{\sqrt{T_0}} \left(\frac{2}{\gamma+1} \right)^{\frac{\gamma}{\gamma-1} - \frac{1}{2}} \frac{A^*}{A}$$

$$\frac{\dot{m}}{A} \sqrt{\frac{R}{\gamma}} = \frac{P_0}{\sqrt{T_0}} \left(\frac{2}{\gamma+1} \right)^{\frac{\gamma+1}{2(\gamma-1)}} \frac{A^*}{A}$$

$$\frac{\dot{m}}{A} \sqrt{\frac{R}{\gamma}} \frac{\sqrt{T_0}}{P_0} = \left(\frac{2}{\gamma+1} \right)^{\frac{\gamma+1}{2(\gamma-1)}} \frac{M}{\left(\frac{2}{\gamma+1} + \frac{\gamma-1}{\gamma+1} M^2 \right)^{\frac{\gamma+1}{2(\gamma-1)}}}$$

$$\frac{\dot{m}}{A} \sqrt{\frac{R}{\gamma}} \frac{\sqrt{T_0}}{P_0} = M \left(\frac{\frac{2}{\gamma+1}}{\frac{2}{\gamma+1} + \frac{\gamma-1}{\gamma+1} M^2} \right)^{\frac{\gamma+1}{2(\gamma-1)}}$$

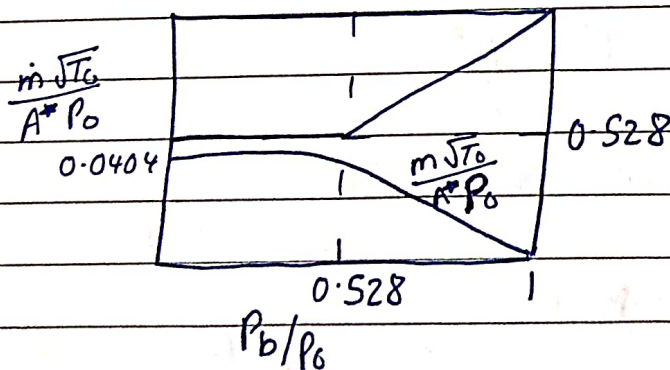
$$\frac{\dot{m}}{A} \sqrt{\frac{R}{\gamma}} \frac{\sqrt{T_0}}{P_0} = M \left(\frac{1}{1 + \frac{\gamma-1}{2} M^2} \right)^{\frac{\gamma+1}{2(\gamma-1)}} \quad \left(\text{Dividing by } \frac{2}{\gamma+1} \right)$$

$$\frac{\dot{m}}{A} \sqrt{\frac{R}{\gamma}} \frac{\sqrt{T_0}}{P_0} = \frac{M}{\left(1 + \frac{\gamma-1}{2} M^2 \right)^{\frac{\gamma+1}{2(\gamma-1)}}}$$

When $M=1$, $A=A^*$, $\dot{m}$ is maximum

$$\frac{\dot{m}_{\max}}{A^*} \sqrt{\frac{R}{\gamma}} \frac{\sqrt{T_0}}{P_0} = \left(\frac{2}{\gamma+1} \right)^{\frac{\gamma+1}{2(\gamma-1)}}$$

Inference: after choking, mass flow rate can be changed only by changing stagnation conditions



For air, $\gamma=1.4$, $R=287 \text{ J/kg K}$

$$\frac{\dot{m}_{\max} \sqrt{T_0}}{A^* P_0} = 0.0404$$

- The air flow from a large vessel in which the pressure is 300 kPa and the temperature is 40°C and

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flows through a nozzle. If the pressure at some section which is identified as section 2 of the discharge nozzle is measured as 200 kPa, find the temperature and velocity at this section. Also, if the Mach number at some other section 3 is 1.5, find the pressure, temperature and velocity assuming the flow to be steady isentropic 1D flow.
(Isentropic - shock free)

Given: $P_0 = 300 \text{ kPa}$ $T_0 = 40^\circ\text{C} = 313 \text{ K}$
 $\gamma = 1.4$ $P_2 = 200 \text{ kPa}$ $M_3 = 1.5$

$$\frac{P_2}{P_{02}} = \frac{2}{3} = 0.6666$$

From Table, $\frac{T_2}{T_{02}} = 0.8915$

$$T_2 = 0.8915 \times 313 = 279 \text{ K}$$

$$\begin{aligned} V_2 &= a_2 M_2 \\ &= \sqrt{\gamma R T_2} \times M_2 \\ &= \sqrt{1.4 \times 287 \times 279} \times 0.78 \\ &= 261 \text{ m/s} \end{aligned}$$

$$M_3 = 1.5$$

From Table, $\frac{P_3}{P_0} = 0.2724$ $\frac{T_3}{T_0} = 0.6897$

$$P_3 = 81.72 \text{ kPa} \quad T_3 = 215.87 \text{ K}$$

$$\begin{aligned} V_3 &= \sqrt{\gamma R T_3} \times M_3 \\ &= 441.766 \text{ m/s} \end{aligned}$$

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2. The velocity, pressure and temperature at a certain point in a steady air flow are 600 m/s, 70 kPa and 5°C respectively. Assuming the flow to be 1D and isentropic, estimate the flow conditions at a point where the pressure is 30 kPa.

Given: $P_1 = 70 \text{ kPa}$ $P_2 = 30 \text{ kPa}$
 $T_1 = 5^\circ\text{C} = 278 \text{ K}$ $V_1 = 600 \text{ m/s}$

$$a_1 = \sqrt{\gamma R T_1} = 334.216 \text{ m/s}$$
$$M_1 = \frac{V_1}{a_1} = 1.798 \approx 1.8$$

From Table, $\frac{P_1}{P_{01}} = 0.174$

$$P_{01} = \frac{70}{0.174} = 402.2988 \text{ kPa}$$

$$P_{01} = P_{02} = 402.2988 \text{ kPa}$$

$$\frac{P_2}{P_{02}} = \frac{30}{402.2988} = 0.0745$$

From Table, $M_2 = 2.34$

$$\frac{T_1}{T_{01}} = 0.6068 \quad (\text{from table})$$

$$T_{01} = \frac{278}{0.6068} = 458.14 = T_{02}$$

From Table, $\frac{T_2}{T_{02}} = 0.4773$

$$T_2 = 218 \text{ K}$$

$$V_2 = \sqrt{\gamma R T_2} \quad M_2 = 692.546 \text{ m/s}$$

3. A gas is contained in a large vessel under a pressure of 300 kPa and temperature of 50°C. The gas is expanded through a nozzle until the flow reaches Mach 1 condition. Estimate the flow properties at this point if the gas is air and if the gas is helium.

Given : $\gamma_{\text{air}} = 1.4$ $\gamma_{\text{helium}} = 1.667$
 $R_{\text{air}} = 287$ $R_{\text{helium}} = 2077$
 $P_0 = 300 \text{ kPa}$ $T_0 = 50^\circ\text{C} = 323 \text{ K}$
 $M_1 = 1$

For air, $\frac{P_1}{P_0} = 0.5283$; $\frac{T_1}{T_0} = 0.8333$

$P^* = P_1 = 158.47 \text{ kPa}$; $T^* = T_1 = 269.1559 \text{ K}$

$V^* = \sqrt{\gamma R T^*}$
 $= 328.856 \text{ m/s}$

For helium,

$\frac{P^*}{P_0} = \left(\frac{2}{\gamma+1} \right)^{\frac{\gamma}{\gamma-1}}$

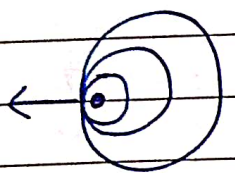
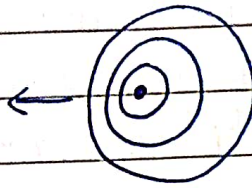
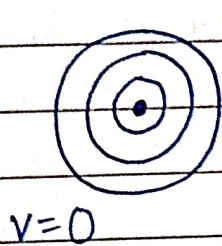
$P^* = 300 \left(\frac{2}{2.667} \right)^{\frac{1.667}{0.667}} = 146.127 \text{ kPa}$

$\frac{T^*}{T_0} = \frac{2}{\gamma+1}$

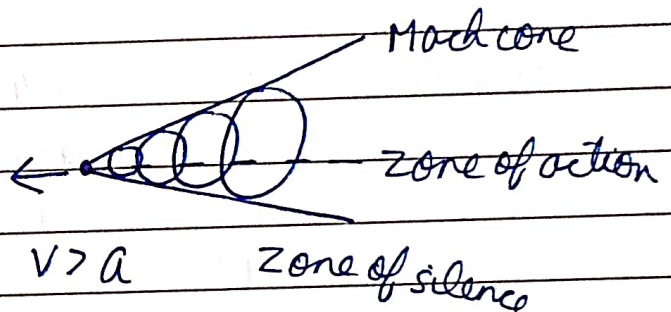
$T^* = \frac{323 \times 2}{2.667} = 242.219 \text{ K}$

$V^* = \sqrt{\gamma R T^*} = 915.7778 \text{ m/s}$

2/4 Introduction to Shock Waves



$v=a$



$v > a$

Propagation of disturbance waves

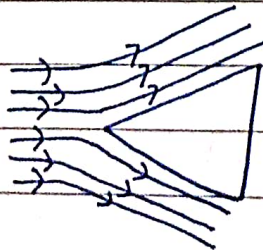
Mach angle

$$\mu = \sin^{-1}\left(\frac{1}{M}\right)$$

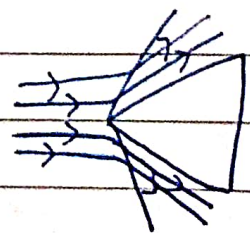
Sound wave: Infinitesimal change in flow properties (isentropic wave)

Shock wave: Variations in properties are very high
 significant change in flow properties

Flow around a wedge

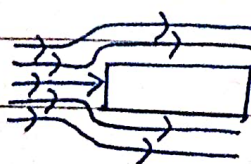


Subsonic

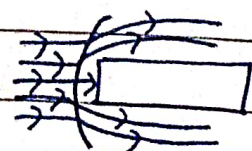


Supersonic

Streamlines around a blunt faced cylinder



Subsonic



Supersonic

Shockwave formation -

When a flow hits a body, it rebounds back and hits the free stream velocity. Coalescence or merging of flow field takes place, both upstream and rebounded flow. The molecules mix together and their random motion causes a shockwave. Here, the tangency of flow is maintained (flow sticks to the wall of the body)

Normal Shockwave: 90°

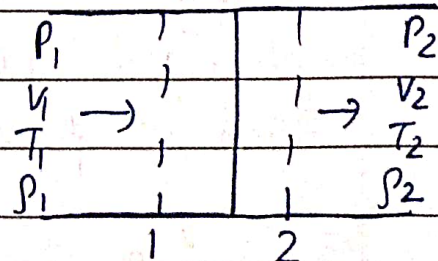
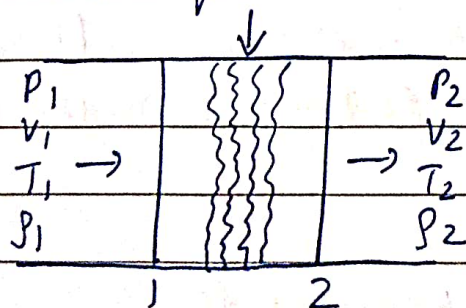
supersonic flow upstream flow is interrupted due to obstruction encountered in the flow field, that is perpendicular to flow direction.

Compression wave

Equations of motion for a normal shock wave

Non-Equilibrium region

shock becomes very thin



effects inside shock region not considered

Consider 2 sections, upstream and downstream of shock
 $\therefore$ shock is contained within the control volume

//_

Consider the shock to be stationary and the air is approaching the shock wave

Across shockwave, Kinetic energy decreases
Internal energy increases
Temperature increases

Continuity eqn: $\rho_1 V_1 = \rho_2 V_2$

Momentum eqn: $P_1 + \rho_1 V_1^2 = P_2 + \rho_2 V_2^2$

Adiabatic Energy eqn: $h_1 + \frac{1}{2} V_1^2 = h_2 + \frac{1}{2} V_2^2$

The flow across a shockwave is adiabatic because no heat transfer happens and the time period for transfer of heat is very less

Is the flow isentropic?

The process is adiabatic but not isentropic due to irreversible nature of the shock wave, the entropy across the shock increases. No heat is exchanged but internal energy increases due to shock compression.

Stagnation temperature remains the same
Stagnation pressure loss - decreased pressure, increased entropy. It is an irreversible process due to viscosity and thermal conductivity that causes losses.

//_

$\therefore$ Across a normal shockwave, $P \uparrow$, $\rho \uparrow$, $T \uparrow$, $M \downarrow$

Effect of back pressure in a CD nozzle

Free shear layer
Shock triple point
Shock barrel
Mach disk

Cases

- 1) Subsonic flow
- 2) Flow just choked
- 3) Shock in nozzle
- 4) Shock at exit
- 5) Overexpanded (oblique shock waves)
- 6) Design condition (shock free)
- 7) Underexpanded

4. A De-Laval Nozzle is to be designed for an exit Mach number of 1.5 with an exit diameter of 200mm. Find the required ratios of the throat area to exit area and the exit flow properties, maximum mass flow rate if the reservoir conditions are 1 atm and 20°C.

Given: $P_0 = 1 \text{ atm} = 101.325 \text{ kPa}$ $T_0 = 20^\circ\text{C} = 293 \text{ K}$
 $M_e = 1.5$ $d_e = 200 \text{ mm}$

According to Table, $\frac{P_e}{P_0} = 0.2724$

$P_e = 27.6 \text{ kPa}$
 $= 0.2724 \text{ atm}$

From Table, $\frac{T_e}{T_0} = 0.6897$

$T_e = 202.08 \text{ K}$

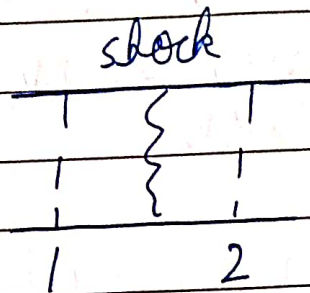
$v_e = \sqrt{\gamma R T_e} M_e = 427.423 \text{ m/s}$

$\frac{A_e}{A^*} = \frac{1}{M} \left(\frac{2}{\gamma+1} + \frac{\gamma-1}{\gamma+1} M^2 \right)^{\frac{\gamma+1}{2(\gamma-1)}}$
 $= \frac{1}{1.5} \left(\frac{2}{2.4} + \frac{0.4 \times 1.5^2}{2.4} \right)^{\frac{2.4}{2 \times 0.4}}$
 $= 1.176$

$\dot{m}_{\max} = \frac{0.0404 \times A^* \times P_0}{\sqrt{T_0}} = \frac{0.0404 \times \pi \times 100^2 \times 101325}{1.176 \times \sqrt{297} \times 10^6} = 6.389 \text{ kg/s}$

±/4 Normal Shock Property Relations

1) Prandtl Meyer Relation (Velocity / Mach No. variation)



As per continuity equation, $\rho_1 A_1 V_1 = \rho_2 A_2 V_2$
 $\rho_1 V_1 = \rho_2 V_2$

Momentum equation: $P_1 + \rho_1 V_1^2 = P_2 + \rho_2 V_2^2$
 $\frac{P_1}{\rho_1 V_1} + \frac{\rho_1 V_1^2}{\rho_1 V_1} = \frac{P_2}{\rho_2 V_2} + \frac{\rho_2 V_2^2}{\rho_2 V_2}$

$$\frac{P_1}{\rho_1 V_1} + V_1 = \frac{P_2}{\rho_2 V_2} + V_2$$

$$a^2 = \gamma R T = \frac{\gamma P}{\rho}$$

$$\frac{a_1^2}{\gamma V_1} + V_1 = \frac{a_2^2}{\gamma V_2} + V_2$$

$$V_2 - V_1 = \frac{a_1^2}{\gamma V_1} - \frac{a_2^2}{\gamma V_2} \quad \text{--- (1)}$$

$$\frac{a_1^2}{\gamma - 1} + \frac{V_1^2}{2} = \frac{a_2^2}{\gamma - 1} + \frac{V_2^2}{2} = \frac{a^{*2}}{2} \left(\frac{\gamma + 1}{\gamma - 1} \right)$$

$$a_1^2 = \left(\frac{\gamma + 1}{2} \right) a^{*2} - \left(\frac{\gamma - 1}{2} \right) V_1^2$$

$$a_2^2 = \left(\frac{\gamma + 1}{2} \right) a^{*2} - \left(\frac{\gamma - 1}{2} \right) V_2^2$$

Substituting a_1^2 and a_2^2 in equation (1),

$$v_2 - v_1 = \left(\frac{\gamma+1}{2}\right) \frac{a^{*2}}{\gamma v_1} - \left(\frac{\gamma-1}{2}\right) \frac{v_1^2}{\gamma v_1} - \left(\frac{\gamma+1}{2}\right) \frac{a^{*2}}{\gamma v_2} + \left(\frac{\gamma-1}{2}\right) \frac{v_2^2}{\gamma v_2}$$

$$v_2 - v_1 = \left(\frac{\gamma+1}{2}\right) \frac{a^{*2}}{\gamma} \left(\frac{1}{v_1} - \frac{1}{v_2}\right) + \left(\frac{\gamma-1}{2}\right) \frac{v_2 - v_1}{\gamma}$$

$$v_2 - v_1 = (v_2 - v_1) a^{*2} \left(\frac{\gamma+1}{2\gamma v_1 v_2}\right) + (v_2 - v_1) \left(\frac{\gamma-1}{2\gamma}\right)$$

Dividing by $v_2 - v_1$,

$$1 = a^{*2} \left(\frac{\gamma+1}{2\gamma v_1 v_2}\right) + \left(\frac{\gamma-1}{2\gamma}\right)$$

$$a^{*2} \left(\frac{\gamma+1}{2\gamma v_1 v_2}\right) = 1 - \frac{\gamma-1}{2\gamma} = \frac{2\gamma - \gamma + 1}{2\gamma}$$

$$a^{*2} \left(\frac{\gamma+1}{v_1 v_2}\right) = \gamma + 1$$

$$\frac{a^{*2}}{v_1 v_2} = 1$$

$$a^{*2} = v_1 v_2$$

$$M^* = \frac{V}{a^*} \Rightarrow \frac{v_1}{a^*} \times \frac{v_2}{a^*} = 1$$

$$M_1^* \times M_2^* = 1$$

we know that $M^* = \frac{(\gamma+1)M^2}{(\gamma-1)M^2+2}$

$$M_1^* = \frac{(\gamma+1)M_1^2}{(\gamma-1)M_1^2+2}, \quad M_2^* = \frac{(\gamma+1)M_2^2}{(\gamma-1)M_2^2+2}$$

Substituting, $\frac{(\gamma+1)M_1^2}{(\gamma-1)M_1^2+2} \times \frac{(\gamma+1)M_2^2}{(\gamma-1)M_2^2+2} = 1$

$$(\gamma+1)^2 M_1^2 M_2^2 = (\gamma-1)^2 M_1^2 M_2^2 + 2(\gamma-1)M_1^2 + 2(\gamma-1)M_2^2 + 4$$

$$M_1^2 M_2^2 [(\gamma+1)^2 - (\gamma-1)^2] = 4 \left[\left(\frac{\gamma+1}{2}\right) M_1^2 + \left(\frac{\gamma-1}{2}\right) M_2^2 + 1 \right]$$

$$M_1^2 M_2^2 \times 4\gamma = 4 \left[\left(\frac{\gamma-1}{2} \right) M_1^2 + \left(\frac{\gamma-1}{2} \right) M_2^2 + 1 \right]$$

$$M_2^2 \left[\gamma M_1^2 - \frac{\gamma-1}{2} \right] = 1 + \left(\frac{\gamma-1}{2} \right) M_1^2$$

$$M_2^2 = \frac{1 + \left(\frac{\gamma-1}{2} \right) M_1^2}{\gamma M_1^2 - \left(\frac{\gamma-1}{2} \right)}$$

Prandtl - Meyer's Relation : $M_2^* = \frac{1}{M_1^*}$

$\Rightarrow$ M_1 - subsonic then M_2 - supersonic
 M_1 - supersonic then M_2 - subsonic

Density Ratio

$$\begin{aligned} \rho_1 V_1 &= \rho_2 V_2 \\ \frac{\rho_2}{\rho_1} &= \frac{V_1}{V_2} = \frac{V_1^2}{V_1 V_2} \\ &= \frac{V_1^2}{a^{*2}} \\ &= M_1^2 \\ &= \frac{(\gamma+1) M_1^2}{(\gamma-1) M_1^2 + 2} \end{aligned}$$

Pressure Ratio

$$\begin{aligned} P_1 + \rho_1 V_1^2 &= P_2 + \rho_2 V_2^2 \\ P_2 - P_1 &= \rho_1 V_1^2 - \rho_2 V_2^2 \\ &= \rho_1 V_1 (V_1 - V_2) \\ &= \rho_1 V_1^2 \left(1 - \frac{V_2}{V_1} \right) \end{aligned}$$

$$\frac{P_2 - P_1}{P_1} = \frac{\rho_1 V_1^2}{P_1} \left(1 - \frac{V_2}{V_1} \right)$$

$$\frac{P_2}{P_1} - 1 = \frac{\gamma \rho_1 V_1^2}{\gamma P_1} \left(1 - \frac{V_2}{V_1} \right)$$

$$\frac{P_2}{P_1} - 1 = \gamma M_1^2 \left(1 - \frac{(\gamma-1)M_1^2 + 2}{(\gamma+1)M_1^2} \right)$$

$$\frac{P_2}{P_1} = 1 + \gamma M_1^2 \left(1 - \frac{(\gamma-1)M_1^2 + 2}{(\gamma+1)M_1^2} \right)$$

$$\frac{P_2}{P_1} = 1 + \gamma M_1^2 \left(\frac{(\gamma+1)M_1^2 - (\gamma-1)M_1^2 - 2}{(\gamma+1)M_1^2} \right)$$

$$\frac{P_2}{P_1} = 1 + \frac{\gamma}{\gamma+1} \left(M_1^2 (\gamma+1 - \gamma+1) - 2 \right)$$

$$\frac{P_2}{P_1} = 1 + \frac{\gamma}{\gamma+1} (2M_1^2 - 2)$$

$$\frac{P_2}{P_1} = 1 + \frac{2\gamma}{\gamma+1} (M_1^2 - 1)$$

$\frac{P_2 - P_1}{P_1} = \frac{\Delta P}{P_1}$ is called the shock strength of a normal shockwave

Temperature Ratio

$$\frac{T_2}{T_1} = \frac{P_2}{P_1} \times \frac{\rho_1}{\rho_2}$$

$$P = \rho RT$$

$$\frac{T_2}{T_1} = \left(1 + \frac{2\gamma}{\gamma+1} (M_1^2 - 1) \right) \times \frac{(\gamma-1)M_1^2 + 2}{(\gamma+1)M_1^2}$$

2) Change in entropy

$$\Delta S = S_2 - S_1 = C_p \ln \left(\frac{T_2}{T_1} \right) - R \ln \left(\frac{P_2}{P_1} \right)$$

$$s_2 - s_1 = C_p \ln \left[\left(1 + \frac{\gamma}{\gamma+1} (M_1^2 - 1) \right) \left(\frac{(\gamma-1)M_1^2 + 2}{(\gamma+1)M_1^2} \right) \right] - R \ln \left[1 + \frac{\gamma}{\gamma+1} (M_1^2 - 1) \right]$$

$\therefore$ change in entropy only depends on upstream Mach number

when $M_1 = 1$, $\Delta s = 0 \Rightarrow$ isentropic

when $M_1 > 1$, $\Delta s > 0 \Rightarrow$ downstream flow is subsonic

when $M_1 < 1$, $\Delta s < 0 \Rightarrow$ not possible for normal shocks

Stagnation pressure ratio (in terms of entropy ratio)

$$T_{01} = T_{02}$$

$$\Delta s = s_2 - s_1 = -R \ln \left(\frac{P_{02}}{P_{01}} \right)$$

$$\frac{-\Delta s}{R} = \ln \left(\frac{P_{02}}{P_{01}} \right)$$

$$\frac{P_{02}}{P_{01}} = e^{-\Delta s/R}$$

5. Calculate the downstream properties across a normal shockwave if the upstream conditions are:

velocity = 680 m/s, pressure = 1 atm, temperature = 300 K

$$V_1 = 680 \text{ m/s}, P_1 = 1 \text{ atm}, T_1 = 300 \text{ K}$$

$$a_1 = \sqrt{\gamma R T_1} = 347.188 \text{ m/s}$$

$$M_1 = \frac{V_1}{a_1} = 1.958 \approx 1.96$$

From Table, $\frac{P_2}{P_1} = 4.3152$

$$\frac{T_2}{T_1} = 1.6553$$

$$P_2 = 4.3152 \text{ atm}$$

$$T_2 = 496.59 \text{ K}$$

$$M_2 = 0.5844$$

$$V_2 = \sqrt{\gamma R T_2} \cdot M_2$$
$$= 295.996 \text{ m/s}$$

9/4 Hugoniot Relation
Properties of shockwave in terms of thermodynamics properties

Continuity equation : $\rho_1 V_1 = \rho_2 V_2$

$$V_2 = \frac{\rho_1 V_1}{\rho_2}$$

$$\Rightarrow P_1 + \rho_1 V_1^2 = P_2 + \rho_2 V_2^2$$
$$P_1 + \rho_1 V_1^2 = P_2 + \rho_2 \left(V_1 \frac{\rho_1}{\rho_2} \right)^2$$

$$V_1^2 \left(\rho_1 - \frac{\rho_1^2}{\rho_2} \right) = P_2 - P_1$$

$$V_1^2 \rho_1 \left(\frac{\rho_2 - \rho_1}{\rho_2} \right) = P_2 - P_1$$

$$V_1^2 = \frac{P_2 - P_1}{\rho_2 - \rho_1} \times \frac{\rho_2}{\rho_1}$$

Similarly, $V_2^2 = \frac{P_2 - P_1}{\rho_2 - \rho_1} \times \frac{\rho_1}{\rho_2}$

Energy equation : $h_1 + \frac{1}{2} V_1^2 = h_2 + \frac{1}{2} V_2^2$

$$H = U + PV, \quad h = e + P v$$

v - specific volume

$$e = \frac{u}{m} \text{ specific internal energy}$$

$$e_1 + p_1 v_1 + \frac{1}{2} v_1^2 = e_2 + p_2 v_2 + \frac{1}{2} v_2^2$$

$$e_1 + p_1 v_1 + \frac{1}{2} \left(\frac{p_2 - p_1}{p_2 - p_1} \right) \left(\frac{p_2}{p_1} \right) = e_2 + p_2 v_2 + \frac{1}{2} \left(\frac{p_2 - p_1}{p_2 - p_1} \right) \left(\frac{p_1}{p_2} \right)$$

$$e_2 - e_1 = p_1 v_1 - p_2 v_2 + \frac{1}{2} \left[\frac{p_2 - p_1}{p_2 - p_1} \right] \left(\frac{p_2}{p_1} - \frac{p_1}{p_2} \right)$$

$$\text{since } p_1 v_1 = \frac{p_1}{p_1} \text{ and } p_2 v_2 = \frac{p_2}{p_2}$$

$$e_2 - e_1 = \frac{p_1}{p_1} - \frac{p_2}{p_2} + \frac{1}{2} \left(\frac{p_2 - p_1}{p_2 - p_1} \right) \left(\frac{p_2^2 - p_1^2}{p_1 p_2} \right)$$

$$e_2 - e_1 = \frac{p_1 p_2 - p_2 p_1}{p_1 p_2} + \frac{1}{2} (p_2 - p_1) \left(\frac{1}{p_1} + \frac{1}{p_2} \right)$$

$$e_2 - e_1 = \frac{p_1 p_2 - p_2 p_1}{p_1 p_2} + \frac{1}{2} \left(\frac{p_2}{p_1} + \frac{p_2}{p_2} - \frac{p_1}{p_1} - \frac{p_1}{p_2} \right)$$

$$e_2 - e_1 = \frac{2p_1 p_2 - 2p_2 p_1 + p_2 p_2 + p_2 p_1 - p_1 p_2 - p_1 p_1}{2 p_1 p_2}$$

$$e_2 - e_1 = \frac{p_1 p_2 - p_2 p_1 - p_1 p_1 + p_2 p_2}{2 p_1 p_2}$$

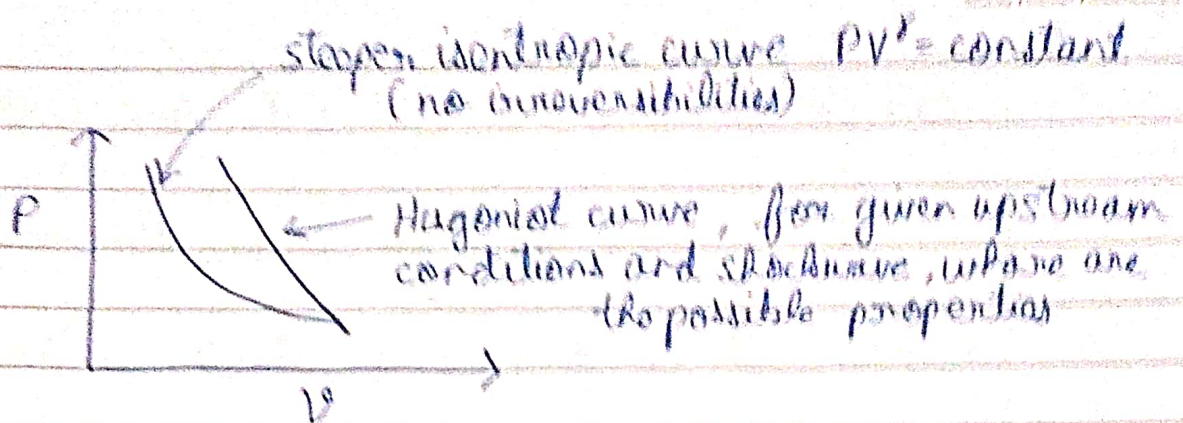
$$e_2 - e_1 = \frac{1}{2} \left(\frac{p_1}{p_1} - \frac{p_2}{p_2} - \frac{p_1}{p_2} + \frac{p_2}{p_1} \right)$$

$$e_2 - e_1 = \frac{1}{2} (p_1 v_1 - p_2 v_2 - p_1 v_2 + p_2 v_1)$$

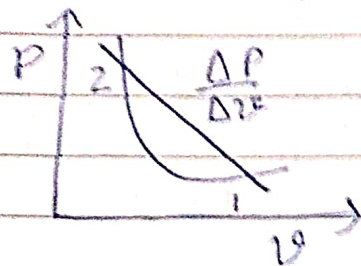
$$e_2 - e_1 = \frac{1}{2} (p_1 + p_2) (v_1 - v_2) \Rightarrow \text{applicable for any perfect gas}$$

$$e = e(p, v)$$

$$p_2 = f(p_1, v_1, v_2)$$



Hugoniot curve for normal shockwave



For given upstream conditions, we can get downstream point 2, so slope of Hugoniot curve can be obtained

$$v_1^2 = \left(\frac{P_2 - P_1}{P_2 - P_1} \right) \left(\frac{P_2}{P_1} \right) \quad \text{or} \quad v_2^2 = \left(\frac{P_2 - P_1}{P_2 - P_1} \right) \left(\frac{P_1}{P_2} \right)$$

$$v_1^2 = \left(\frac{P_2 - P_1}{\frac{1}{v_2} - \frac{1}{v_1}} \right) \left(\frac{v_1}{v_2} \right) = \frac{P_2 - P_1}{\frac{v_1 - v_2}{v_1 v_2}} \left(\frac{v_1}{v_2} \right)$$

$$v_1^2 = v_1^2 \left(\frac{P_2 - P_1}{v_1 - v_2} \right)$$

$$\frac{P_2 - P_1}{v_2 - v_1} = - \frac{v_1^2}{v_1^2} \Rightarrow \text{slope of Hugoniot curve}$$

$$e = C_v T = \frac{R T}{\gamma - 1} = \frac{P v}{\gamma - 1}$$

$$e_2 - e_1 = \frac{P_1 + P_2}{2} (v_1 - v_2)$$

$$\frac{P_2 v_2 - P_1 v_1}{\gamma - 1} = \frac{P_1 + P_2}{2} (v_1 - v_2)$$

$$\frac{P_2 v_2 - P_1 v_1}{v_1 - v_2} = \frac{P_1 + P_2}{2} (\gamma - 1)$$

$$P_2 \left(\frac{v_2}{v_1 - v_2} \right) - P_1 \left(\frac{v_1}{v_1 - v_2} \right) = P_1 \left(\frac{\gamma - 1}{2} \right) + P_2 \left(\frac{\gamma - 1}{2} \right)$$

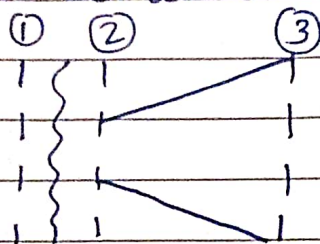
$$P_2 \left[\frac{v_2}{v_1 - v_2} - \frac{\gamma - 1}{2} \right] = P_1 \left[\frac{v_1}{v_1 - v_2} + \frac{\gamma - 1}{2} \right]$$

$$\frac{P_2}{P_1} = \frac{\frac{v_1}{v_1 - v_2} + \frac{\gamma - 1}{2}}{\frac{v_2}{v_1 - v_2} - \frac{\gamma - 1}{2}}$$

$$\frac{P_2}{P_1} = \frac{\frac{\gamma + 1}{\gamma - 1} \left(\frac{v_1}{v_2} \right) - 1}{\frac{\gamma + 1}{\gamma - 1} - \frac{v_1}{v_2}}$$

6. The ratio of exit to entry area in a subsonic diffuser is 4. The Mach number of the jet of air approaching the diffuser is at 2.2 and $P_0 = 1.013 \text{ bar}$ and $T_0 = 290 \text{ K}$. There is a standing normal shockwave at the entry of the diffuser and if the flow through the diffuser is isentropic, determine the exit conditions of the diffuser

11/4



$$A_3/A_2 = 4$$

$$P_{01} = 1.013 \text{ bar}$$

$$T_1 = 290 \text{ K}$$

$$M_1 = 2.2$$

$M_1 = 2.2$, From table,

$$\frac{P_1}{P_{01}} = 0.0935$$

$$\frac{T_1}{T_{01}} = 0.5081$$

$$P_1 = 0.0947 \text{ bar}$$

$$T_{01} = 570.75 \text{ K}$$

$M_1 = 2.2$, From table,

$$M_2 = 0.5471$$

$$\frac{P_2}{P_1} = 5.48$$

$$\frac{T_2}{T_1} = 1.8569$$

$$P_2 = 0.5189 \text{ bar}$$

$$T_2 = 538.5 \text{ K}$$

$M_2 = 0.5471$, From table, $\frac{A}{A^*} = 1.26$

$$A_2^* = A_3^*$$

$$\frac{A_3}{A_3^*} = \frac{A_3}{A_2} \times \frac{A_2}{A_2^*} = \frac{A_3}{A_2} \times \frac{A_2}{A_2^*}$$

$$= 4 \times 1.26$$

$$= 5.04$$

From Table, $M_3 = 0.116$

Normal shock, $\frac{P_{02}}{P_{01}} = 0.6281$

$$P_{02} = 0.6363 \text{ bar}$$

$$P_{03} = P_{02} = 0.6363 \text{ bar}$$

$$T_{03} = T_{02} = T_{01} = \cancel{290 \text{ K}} \\ 570.75 \text{ K}$$

$M_3 = 0.116$, From table,

$$\frac{P_3}{P_{03}} = 0.991$$

$$\frac{T_3}{T_{03}} = 0.9975$$

$$P_3 = 0.63 \text{ bar}$$

$$T_3 = 569.323 \text{ K}$$

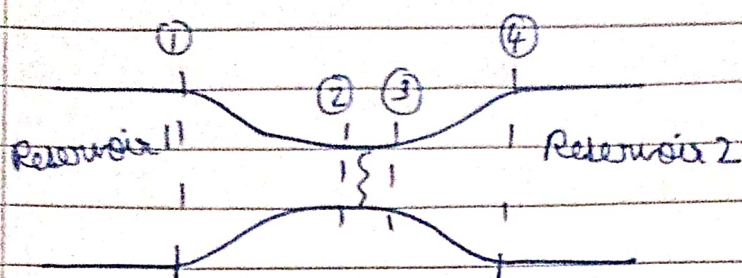
Stagnation Pressure loss $P_{03} - P_{01} = 0.63 - 1.013$
 $= 0.383 \text{ bar}$

7. Air is expanded from a large reservoir in which the pressure and temperature are 500 kPa and 35°C, through a variable area duct. A normal shock occurs at a point / section in the duct where the mach number is 2.5. Find the flow properties after the shock. The downstream flow is brought to rest in another reservoir kept after the duct. Find the conditions of air in this reservoir. Assume Quasi 1D flow and the flow is isentropic except across the shockwave.

$$P_{01} = 500 \text{ kPa}$$

$$T_{01} = 35^\circ\text{C} = 308 \text{ K}$$

$$M_2 = 2.5$$



$$P_{02} = P_{01} = 500 \text{ kPa}$$

$$T_{01} = T_{02} = T_{03} = T_{04} = 308 \text{ K}$$

$$P_{03} = P_{04}$$

At Section 2, $M_2 = 2.5$, From Table

$$\frac{P_2}{P_{02}} = 0.0585$$

$$\frac{T_2}{T_{02}} = 0.4444$$

$$P_2 = 29.25 \text{ kPa}$$

$$T_2 = 136.8752 \text{ K}$$

At Section 3, $M_3 = 2.5$, From Table

$$M_3 = 0.512$$

$$\frac{P_3}{P_2} = 7.125$$

$$\frac{T_3}{T_2} = 2.1375$$

$$P_3 = 208.406 \text{ kPa}$$

$$T_3 = 292.5 \text{ K}$$

$$\frac{P_{03}}{P_{02}} = 0.499$$

$$P_{03} = 249.5 \text{ kPa}$$

At Section 4, $P_{04} = P_{03} = 249.5 \text{ kPa}$

$$T_{04} = 308 \text{ K}$$

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Impulse Function

For a steady 1D compressible flow, the impulse function is given by $F = pA + \rho u^2 A$ (Net force exerted by a flow and is

Isentropic flow $F = pA(1 + \gamma M^2)$ derived from momentum equation)
for an ideal gas

Physical significance

- Represents momentum flux plus pressure force

Non dimensional representation of impulse function

$$\frac{F}{P_0 A^*} = \frac{1 + \gamma M^2}{M} \sqrt{\left(\frac{2}{\gamma + 1} \left(1 + \frac{\gamma - 1}{2} M^2 \right) \right)^{\frac{\gamma + 1}{\gamma - 1}}}$$

This uses $\frac{P}{P_0} = \left(1 + \frac{\gamma - 1}{2} M^2 \right)^{\frac{-\gamma}{\gamma - 1}}$

$$\frac{A}{A^*} = \frac{1}{M} \left(\frac{2}{\gamma + 1} \left(1 + \frac{\gamma - 1}{2} M^2 \right) \right)^{\frac{\gamma + 1}{2(\gamma - 1)}}$$

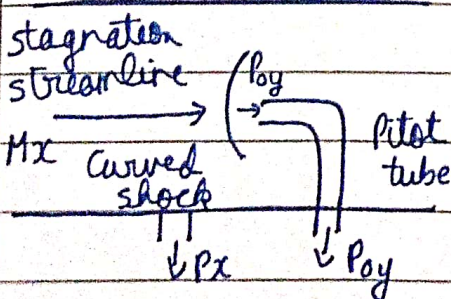
Another form: $\frac{F}{F^*} = \frac{P}{P^*} \cdot \frac{A}{A^*} \cdot \frac{1 + \gamma M^2}{1 + \gamma}$

$$= \frac{1 + \gamma M^2}{M \sqrt{2(1 + \gamma) \left(1 + \frac{\gamma - 1}{2} M^2 \right)}}$$

Rayleigh's supersonic pitot formula

relates downstream pressure to upstream pressure of the shockwave

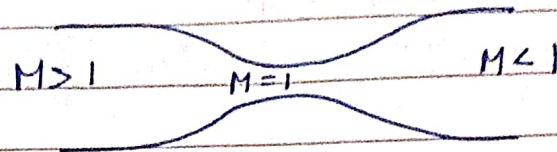
$$\frac{P_{0y}}{P_x} = \frac{\left(\frac{\gamma + 1}{2} M_x^2 \right)^{\frac{\gamma}{\gamma - 1}}}{\left(\frac{2\gamma}{\gamma + 1} M_x^2 - \frac{\gamma - 1}{\gamma + 1} \right)^{\frac{1}{\gamma - 1}}}$$



Stagnation streamline: it is a dividing streamline between the streamlines going up and down

Diffuser

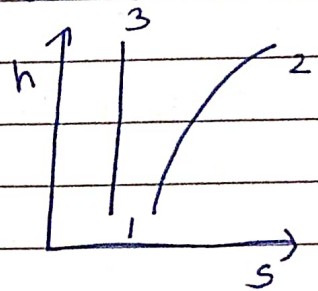
To reduce speed,
increase pressure
 $\Rightarrow$ compression



The diffusion process occurs through compression.
In a supersonic flow, shockwaves give the
compression effect

Diffusion $\rightarrow$ compression $\rightarrow$ change in enthalpy
temperature increases

Isentropic, reversible adiabatic
process: $h_1 \neq h_3$; $h_{01} = h_{03}$



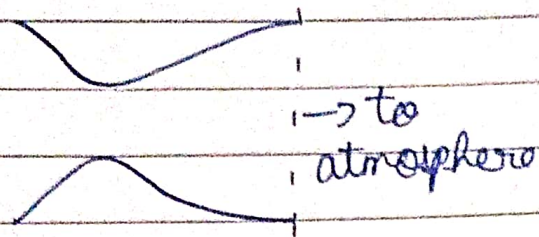
Non-isentropic: $h_1 \rightarrow h_2$

Diffuser efficiency $\eta_D = \frac{\text{change of enthalpy in reversible diffuser}}{\text{change of enthalpy in actual diffuser}}$

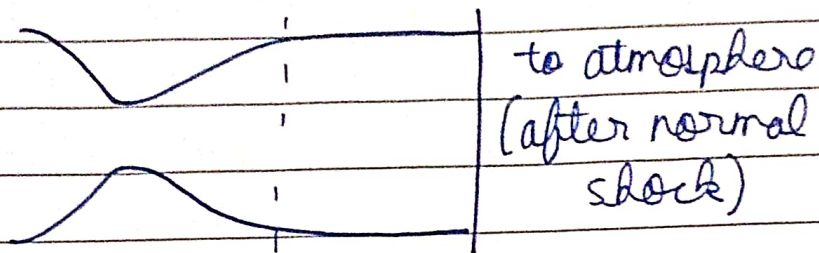
$$\eta_D = \frac{h_{2s} - h_1}{h_2 - h_1}$$

$$\eta_D = \frac{\frac{\gamma+1}{\gamma-1}}{\left(\frac{2\gamma}{\gamma+1} M_1^2 - \frac{\gamma-1}{\gamma+1}\right)^{1/\gamma}} - \left(\frac{2}{\gamma-1}\right) \frac{1}{M_1^2}$$

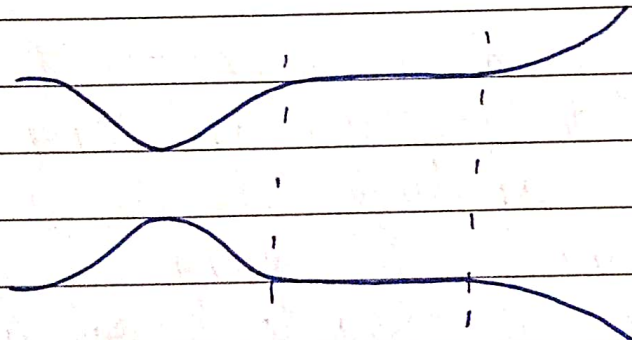
Advantage of Diffuser



Normal shock diffuser



by adding an extension to the nozzle, the reservoir stagnation pressure can be reduced.



Add another section after the normal shock
(subsonic diffuser)

The exit pressure is almost atmospheric pressure and stagnation pressure will be reduced even more. This results in a very high pressure loss across the entire diffuser, which will decrease its efficiency. Thus, design is enhanced using oblique shockwaves.

Drawbacks

-
-
-
-
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-
-

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8. Air is expanded through a CD nozzle from a large reservoir in which the pressure and temperature are 600 kPa and 40°C respectively. The design back pressure is 100 kPa. Determine the following:
- (i) Ratio of nozzle exit area to throat area
 - (ii) The discharge velocity from the nozzle for the design condition
 - (iii) What is the back pressure for which there will be a normal shock at the exit plane of the nozzle?

$$P_0 = 600 \text{ kPa} \quad T_0 = 40^\circ\text{C} = 313 \text{ K}$$

$$P_b = 100 \text{ kPa}$$

$$(i) \frac{P_e}{P_0} = \frac{1}{6} = 0.1667$$

$$\text{From table, } \frac{A}{A^*} = 1.4723 \quad M_e = 1.83$$

$$(ii) \frac{T_e}{T_0} = 0.5989$$

$$T_e = 187.455 \text{ K}$$

$$V_e = \sqrt{\gamma R T_e} \times M_e$$

$$= 502.232 \text{ m/s}$$

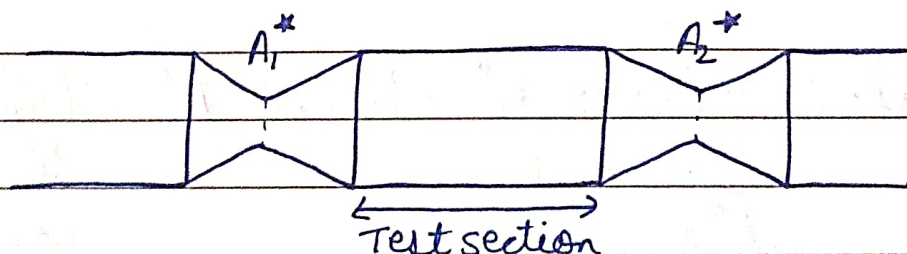
$$(iii) P_b = P_e = \text{Normal shock exit}$$

$$M_e = 1.83$$

$$\text{From table, } M = 0.615$$

$$\frac{P}{P_e} = 3.7405$$

$$P = 374.05 \text{ kPa}$$



To move the normal shock from the test section to the diffuser throat, we increase the throat area of the diffuser to ideal conditions (by reducing pressure ratio) Area of diffuser should allow flow through after shock is formed

$$P_1 A_1^* V_1^* = P_2 A_2^* V_2^*$$

$$V_1^* = a_1^* = a^* = a_2^* = V_2^*$$

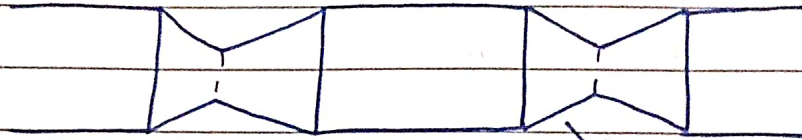
$$P^* = \frac{P}{RT^*} \quad \text{and} \quad T_1^* = T_2^*$$

$$\therefore \frac{A_{t2}}{A_{01}} = \frac{P_1^*}{P_2^*}$$

$$\frac{A_y^*}{A_x^*} = \frac{P_{0x}}{P_{0y}} > 1$$

Case 1: No shock in the whole tube $\Rightarrow$ ideal

Case 2: Shock comes $\Rightarrow$ can't have it in test section as we want supersonic flow there, so we need to push it to the diffuser. Optimum location is the throat.



$\rightarrow$ In this section, supersonic flow is there, so oblique shockwaves are formed

Consider Diffuser $A^* >$ Nozzle A^*

$$\frac{A_y^*}{A_x^*} = \frac{P_{0x}}{P_{0y}} > 1$$

(if opposite, the shockwave comes in the test section and no proper supersonic flow is established)

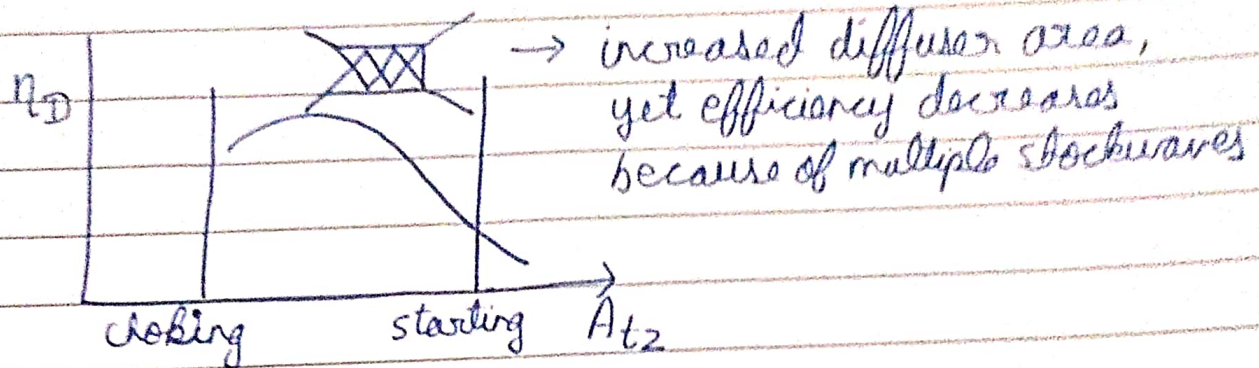
Diffuser efficiency (previous η_D expression is in terms of thermodynamics)

$$\eta_D = \frac{(P_{d0}/P_0)_{\text{actual}}}{(P_{02}/P_{01})_{\text{normal shock at } M_2}}$$

P_{d0} at exit of diffuser after the diffuser

1 / 1

Actual total pressure ratio - total pressure ratio across the hypothetical normal shockwave at the test section Mach no. - P_{02}/P_{01}



9. A CD diffuser is to be used for a $M=3$ supersonic flow. Diffuser has to use a variable throat area so as to swallow the starting shock. What percentage increase in throat area is necessary?

$$\frac{A_{t2}^*}{A_{t1}^*} = \frac{P_{0x}}{P_{0y}} = 0.3283$$

% increase $\Rightarrow$

Let initial $A_{t1}^* = A_{t2}^*$

$$\frac{A_{t2}^* - A_{t1}^*}{A_{t1}^*} = 32.8\%$$

$$= 1 - 0.3283$$

$$= 67.17\%$$